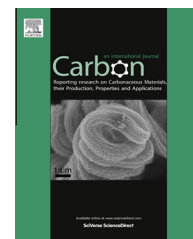


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EMI shielding effectiveness of carbon based nanostructured polymeric materials: A comparative study

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ABSTRACT

The microstructure, electromagnetic interference (EMI) shielding effectiveness (SE), DC electrical conductivity, AC electrical conductivity and complex permittivity of nanostructured polymeric materials filled with three different carbon nanofillers of different structures and intrinsic electrical properties were investigated. The nanofillers were multiwall carbon nanotubes (MWCNT), carbon nanofibers (CNF) and high structure carbon black (HS-CB) nanoparticles and the polymer was acrylonitrile-butadiene-styrene (ABS). In addition, the EMI SE mechanisms and the relation between the AC electrical conductivity in the X-band frequency range and the DC electrical conductivity were studied. The nanocomposites were fabricated by solution mixing and characterized by uniform dispersion of the nanofillers within the polymer matrix. It was found that, at the same nanofiller loading, the EMI SE, permittivity and electrical conductivity of the nanocomposites decreased in the following order: MWCNT > CNF > CB. MWCNT based nanocomposites exhibited the lowest electrical percolation threshold and the highest EMI SE owing to the higher aspect ratio and electrical conductivity of MWCNT compared to CNF and HS-CB. The AC conductivity in the X-band frequency range was found to be independent of frequency.

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1. Introduction

Because of their low density, design flexibility, ease of processing and high conductivity at low filler loading, polymer nanocomposites based on high aspect ratio conductive nanofillers are promising advanced materials for protection from electromagnetic interference (EMI) in laptops, cell phones, aircraft electronics, military and medical devices [1–6]. Recently, the EMI shielding capabilities of polymeric materials filled with high aspect ratio conductive nanofillers, such as single-walled carbon nanotubes (SWCNTs) [3,6–10], multi-walled carbon nanotubes (MWCNTs) [11–18], carbon nanofibers (CNFs) [19–22] and metal nanowires (MNWs) [23,24], have been investigated. It was evident that polymers filled with high aspect ratio nanofillers exhibit higher EMI shielding

effectiveness (SE) than polymers filled with the conventional microfillers such as carbon fiber (CF) [19].

However, it is very difficult to compare and evaluate the EMI SE performance of the nanocomposite materials based on the literature data because the EMI SE data varies with many uncontrolled factors such as the thickness of the shielding samples, the frequency at which the EMI shielding performance was carried out, the polymer matrix type, the fabrication method and processing conditions. Thus, it is of the major objectives of this work to provide a direct comparison between polymer nanocomposites based on three different carbon nanofillers, namely: high structure carbon black (HS-CB), CNF, and MWCNT. The nanocomposites were prepared under identical conditions using acrylonitrile-butadiene-styrene (ABS) as the polymer matrix, solution processing was used

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for fabrication and specimens of 1.1 mm in thickness were used for EMI SE characterization. Detailed information about the specific characteristics of the nanofillers used in the study can be found in the experimental section.

Carbon black (CB) nanoparticles are amorphous type of carbon produced on a large scale by five different processes [25]. CB nanoparticles used as filler in polymers are mainly produced by thermal decomposition of oil [25]. During the manufacturing process, CB prime particles of sizes in the range of 10–100 nm are fused to form CB aggregates. The size and shape of CB aggregate in addition to the number of prime CB particles per aggregate determines the structural classification of CB [25]. CB aggregates that are characterized by long and irregular branches composed of many prime particles are known as HS-CB. While aggregates that contain less branching and fewer number of prime particles are termed low structure carbon black (LS-CB). The irregular and long chains of the HS-CB allow the aggregates to form conductive networks at lower nanofiller loading compared to the low structure CB. The typical resistivity of HS-CB is $\sim 10^{-2} \Omega\text{-cm}$.

CNFs are high aspect ratio nanomaterials of hollow core structure produced by decomposition of a hydrocarbon feedstock or carbon monoxide on a metal catalyst [26,27]. The single nanofiber is formed of stacked graphene layers that are nested with each other and have different structures including bamboo-like, parallel and cup-stacked. CNF have typical diameter is in the range of 50–200 nm and a length of up to 100 μm . As-grown CNF are typically formed of disordered graphene planes. To enhance the electrical conductivity of the nanofibers, CNF are heat treated. The heat treatment of CNF enhances the nanofibers crystallinity, and consequently the electrical conductivity, by re-arranging the carbon planes and removing the amorphous carbon deposited on the surface of the nanofibers [28]. Electrical conductivity of $10^{-4} \Omega\text{-cm}$ was reported for graphitized CNF [19].

The last and most promising nanofiller is CNT. CNT have higher aspect ratio and smaller diameter compared to CNF. Thus, the electrical percolation threshold concentration of CNT based nanocomposites is lower than that of CNF based nanocomposites. There are many processes to manufacture CNT including: laser ablation, arc discharge and chemical vapor deposition (CVD) [29,30]. Among these processes, CVD is the most promising one because of its simplicity that might allow continuous production of large amount of CNT at relatively low cost [29,30]. The purity, size and chirality of CNT determine its electrical conductivity [31]. Multiwall carbon nanotubes (MWCNTs), which are the CNT type investigated in this study, consist of several concentric cylindrical graphene layers that bonded with each other by Van der Waals forces [32]. The diameter of MWCNT is typically in the range of 5–50 nm. Electrical resistivity in the range of 10^{-3} – $10^{-5} \Omega\text{-cm}$ was reported for MWCNT [32].

2. Experimental

2.1. Materials

The ABS copolymer (Cyclocac EX39, specific gravity = 1.03, MFI = 1 g/10 min) used in this study was obtained from SABIC Innovative Plastic. Multiwall carbon nanotubes (MWCNT)

were purchased from Nanocyl S.A. (Nanocyl™ NC7000 nanotubes, Sambreville, Belgium). According to the manufacturer, Nanocyl™ NC7000 has an average diameter of 9.5 nm, a length of 1.5 μm , carbon purity of 90% and surface area of 250–300 m^2/g . Recent study by Castillo and coworkers [33] showed that Nanocyl™ NC7000 nanotubes have a mean diameter and length of 10.0 nm and 1.34 μm , respectively. The HS-CB was kindly provided by Cabot (BP2000, surface area 1487 m^2/g , and prime particle size 12 nm [34]). The CNF were low heat treated (LHT)-Pyrograf III™ carbon nanofibers (PR-24-LHT) produced by Applied Sciences, Inc., Ohio-USA. Recently, we have investigated the diameter and length of PR-24-LHT CNF [20]. We found that PR-24-LHT has an average diameter of $110 \pm 27 \text{ nm}$ and a number average length of $4.2 \pm 4.0 \mu\text{m}$ [20]. HPLC grade chloroform was used for the nanocomposites preparation and nanofiller dispersion.

2.2. Nanocomposites preparation

All nanocomposites were prepared by solution processing according the following procedure. Two grams of ABS pellets were dissolved in 100 mL chloroform. Different amounts of nanofillers, depending on the required mass fraction of the nanofiller in the final nanocomposite, were suspended in 40 mL chloroform by sonication for 10 min. The nanotubes suspension was added to ABS solution all at once and the mixture was sonicated for another 10 min. The mixture was cast at room temperature on glass dish and left overnight to evaporate the solvent. Then the samples were placed in a vacuum oven for three hours at 70 °C to remove the remaining traces of solvent. For the electrical conductivity, dielectric and EMI SE characterization, 1.1 mm thick rectangular samples ($4 \times 2 \text{ cm}^2$) were prepared by compression molding using Carver hot press (Carver Inc., Wabash-IN, USA) at 250 °C under 5 T pressure for 10 min.

2.3. Characterization

The dispersion of the nanofillers was investigated by transmission electron microscopy (TEM). Thin sections $\sim 80 \text{ nm}$ in thickness were cut from the molded specimens using Reichert-Jung Ultracut E ultramicrotome. The TEM micrographs were collected at different magnifications using Zeiss EMIOCR microscope at an acceleration voltage of 80 kV.

The DC electrical resistivity of the nanocomposites was conducted using two different setups. For nanocomposites with electrical conductivity $< 10^{-6} \Omega\text{cm}$, a set-up consisting of Keithley 6517B electrometer (Keithley, Ohio, USA) and a Keithley 8009 test fixture was used. For more conductive nanocomposites, Loresta GP (MCP-T610 model, Mitsubishi Chemical Co., Japan) electrical resistivity meter connected to a four-pin probe (MCP-TP08P model, Mitsubishi Chemical Co., Japan) was used. The probe's pin diameter is 2 mm and the inter-pin spacing is 5 mm. For the EMI SE characterization in the X-band (8.2–12.4 GHz) frequency, ENA series network analyzer (Model E5071C) connected with a WR-90 rectangular waveguide was used. The rectangular specimens were placed between the two sections of the waveguide and the S-parameters (S_{11} , S_{12} , S_{22} , S_{21}) of each sample were recorded over the X-band frequency range. Then the total SE, shielding by

reflection, shielding by absorption, AC conductivity and complex permittivity were calculated. A MATLAB code based on Nicolson–Ross–Weir (NRW) method was developed to extract the complex permittivity from the s-parameters [35,36]. For each formulation, at least three specimens were fabricated and tested.

3. Results and discussion

3.1. Microstructure

TEM was used to investigate the dispersion state of the nanofillers in the ABS polymer matrix. Fig. 1 depicts TEM micrographs of 5 wt.% nanofiller/ABS nanocomposite. General speaking, regardless of the nanofiller type, good level of nanofiller dispersion can be observed. The proper dispersion of the nanofillers within the polymer can be mainly related to the processing method used to prepare the nanocomposite (low viscosity medium with sonication), the selection of a proper solvent (in this work chloroform) to pre-disperse the nanofillers and the compatibility between the nanofillers and the ABS polymer matrix. Micrographs (b) and (c) clearly show, respectively, CNF and CNT particles forming networks within the polymer matrix. It is very difficult to observe networks of CB aggregates, in micrograph (a), because of the low aspect ratio of the aggregate.

3.2. Electrical properties

Fig. 2 depicts the DC electrical resistivity (ρ) of the CB/ABS, CNF/ABS and CNT/ABS nanocomposites as function of the nanofiller weight percent. In general, all nanocomposites exhibited a decrease in the electrical resistivity with the increase in nanofiller concentration. This observation is theoretically expected since increasing the nanofiller concentration creates more networks within the insulating ABS polymer matrix and consequently reduces the nanocomposite overall electrical resistivity. The electrical resistivity of nanocomposites is also a function of the nanofiller aspect ratio and intrinsic electrical conductivity. CNT/ABS nanocomposite has the lowest electrical percolation threshold concentration, which is the critical concentration required for a conductive network to be formed within the polymer matrix. It is apparent that the electrical percolation threshold is between 4.0 and 5.0 wt.% for the CB/ABS nanocomposites, between 1.25 and 1.5 wt.% for the CNF/ABS nanocomposite and below 0.5 wt.% for the CNT/ABS nanocomposite. At 0.5 wt.%, the CNT/ABS nanocomposite is conductive; while at the same loading the CB and CNF based nanocomposites are not. CNT has higher aspect ratio (estimated ~ 300 [37]) than CNF (estimated ~ 38 [20]) and the HS-CB nanoparticles that have irregular-branched structure. The high aspect ratio facilitates the nanofiller/nanofiller conductive network formation at very low nanofiller loading.

The influence of the nanofiller intrinsic conductivity on the nanocomposite bulk conductivity is apparent above the electrical percolation threshold, where at the same nanofiller loading, the electrical resistivity of the nanocomposites decreases with the decrease in nanofiller intrinsic conductivity.

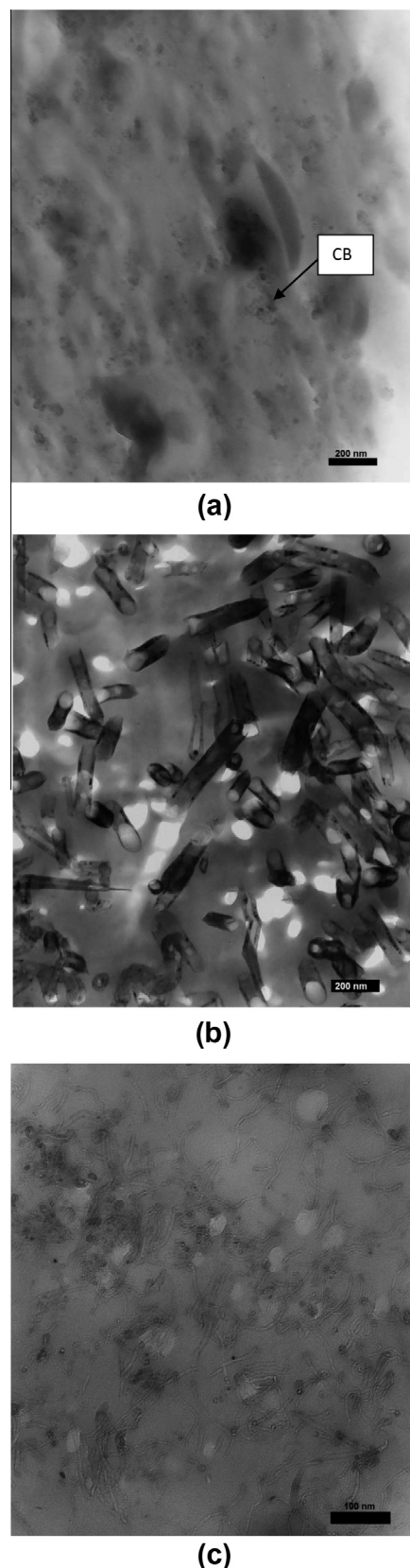


Fig. 1 – TEM micrograph of 5 wt.% nanofiller/ABS nanocomposites: (a) CB, (b) CNF and (c) CNT. Scale bar is 200 nm in micrographs a and b and 100 nm in micrograph c.

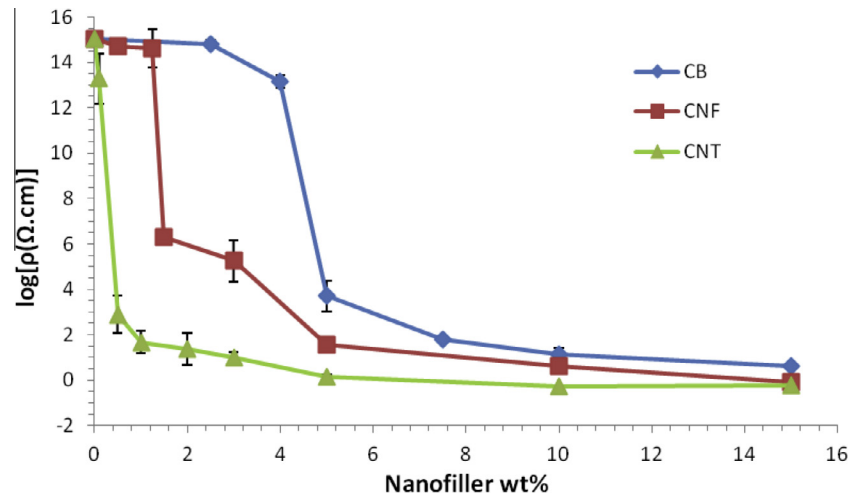


Fig. 2 – Electrical volume resistivity of CB/ABS, CNF/ABS and CNT/ABS nanocomposites as function of nanofiller content. Error bars represents the standard deviation.

Even though it is technically difficult to directly measure the electrical conductivity of most of the nanofillers because of their small sizes, the available estimates suggest that the electrical conductivity of CNT and CNF are higher than that of CB [19]. In addition, based on the electrical resistivity results obtained in this work, we may conclude that the electrical conductivity of CNT is higher than that of CNF, since at very high loadings (15 wt.% in this study) the electrical resistivity of the CNT/ABS nanocomposite ($0.81 \pm 0.05 \text{ Ohm-cm}$) was lower than that of CNF/ABS nanocomposite ($1.5 \pm 0.1 \text{ Ohm-cm}$).

3.3. Electromagnetic Interference Shielding Effectiveness

The overall EMI SE was calculated based on the S-parameters as follows:

$$\text{EMI SE} = 10 \log \frac{1}{|S_{12}|^2} = 10 \log \frac{1}{|S_{21}|^2} \quad (1)$$

where $|S_{ij}|^2$ represents the power transmitted from port i to port j . Fig. 3(a–c) show, respectively, the EMI SE of the CB/ABS, CNF/ABS and CNT/ABS nanocomposites as a function of nanofiller content in the X-band frequency range. The results reported in Fig. 3 are for 1.1 mm thick plates. Most of the nanocomposites showed marginal dependence or even independence of EMI SE on the frequency; similar behavior has been reported for various types of polymer nanocomposites [12,13,24,38]. For example, for the 10 wt.% CNT/ABS nanocomposite, the maximum and minimum EMI SE values that were reported in this study over the X-band frequency range were 40.7 dB and 39.9 dB, respectively. For the CB/ABS nanocomposite (especially the 10 wt.% and 15 wt.% nanocomposites), the EMI SE versus frequency curve, as shown in Fig. 3a, exhibited a wave-like behavior. Basuli et al. [13] reported identical pattern for CNT/ethylene–methyl–acrylate nanocomposites and ascribed such behavior to the irregular nature of the conductive networks.

Fig. 4 shows the influence of nanofiller type and nanofiller content on the EMI SE of ABS nanocomposites at 8.2 GHz. First

of all, for the same nanocomposite, it is apparent that the EMI SE increases with the increase in nanofiller content. For instance, the EMI SE of CB/ABS nanocomposite increased from 8.2 dB to 21.1 dB with increasing CB concentration from 7.5 wt.% to 15 wt.%. Secondly, at the same nanofiller loading, the EMI SE of CNT based nanocomposites is significantly higher than that of CNF and CB based nanocomposites. At 10 wt.% nanofiller loading, the EMI SE for the CNT, CNF and CB nanocomposites were 40.7 dB, 26.1 dB and 15.3 dB, respectively. These numbers demonstrate the superior EMI shielding performance of the CNT based materials compared to the CNF and HS-CB based nanocomposites. Finally, the amount of CNT required to formulate a nanocomposite with EMI SE of 20 dB (a SE level suitable for many applications [19,38]) is only about 2 wt.%. However, in order to attain the same level of shielding, a 15 wt.% CB/ABS nanocomposite is required (corresponding to 7.5 times the amount of CNT).

3.4. Shielding mechanisms

EMI is attenuated by three major mechanisms, namely: reflection, absorption and multiple reflections. In cases where the shielding by absorption (i.e. absorption loss) is higher than 10 dB, most of the re-reflected wave will be absorbed within the shield [39,40]. Thus, multiple reflections can be ignored. In a previous work [11,41], we found that contribution of absorption loss to the total shielding is higher than the contribution of reflection loss. Similar finding was reported by many researchers for different nanocomposite materials [7,12,38,42]. In this work, it was also found that absorption loss was higher than the reflection loss. On average, the absorption loss contributes to 75% of the total EMI SE. As a representative example, the power balance and the reflection loss and absorption loss of CNF/ABS nanocomposites are presented in Fig. 5. Total shielding (SE_T), absorption loss (SE_A), reflection loss (SE_R), reflected power (R), transmitted power (T) and absorbed (A) power were calculated based on the S-parameters obtained from the vector network analyzer as follows:

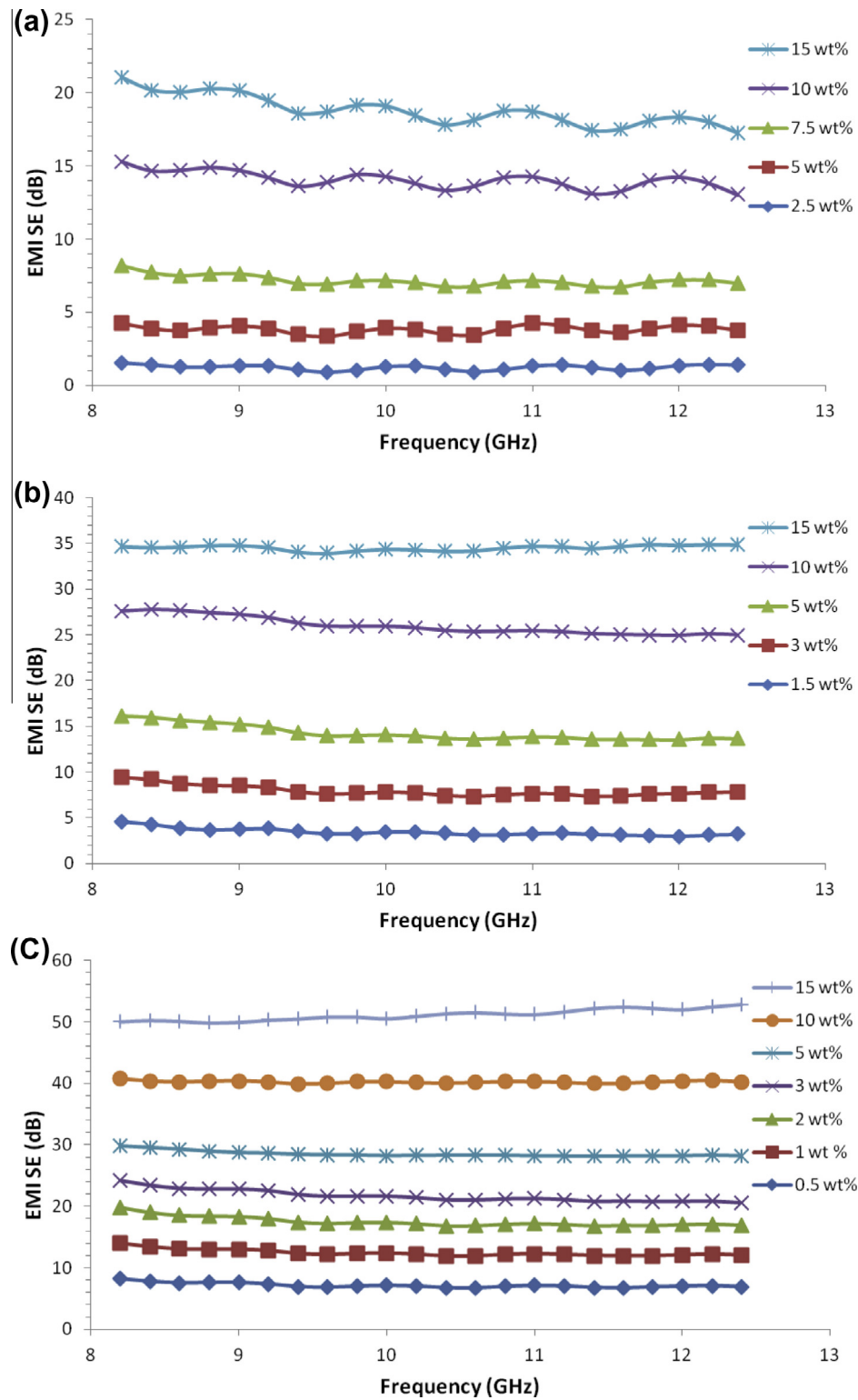


Fig. 3 – EMI SE in the X-band frequency range for 1.1 mm thick plates made from (a) CB/ABS nanocomposite, (b) CNF/ABS nanocomposite and (c) CNT/ABS nanocomposite.

$$SE_T = SE_R + SE_A$$

$$(2) \quad R = |S_{11}|^2 = |S_{22}|^2 \quad (5)$$

$$SE_R = 10 \log_{10}(1/(1 - |S_{11}|^2))$$

$$(3) \quad T = |S_{12}|^2 = |S_{21}|^2 \quad (6)$$

$$SE_A = 10 \log_{10} \left(\frac{1 - |S_{11}|^2}{|S_{12}|^2} \right)$$

$$(4) \quad A = 1 - R - T \quad (7)$$

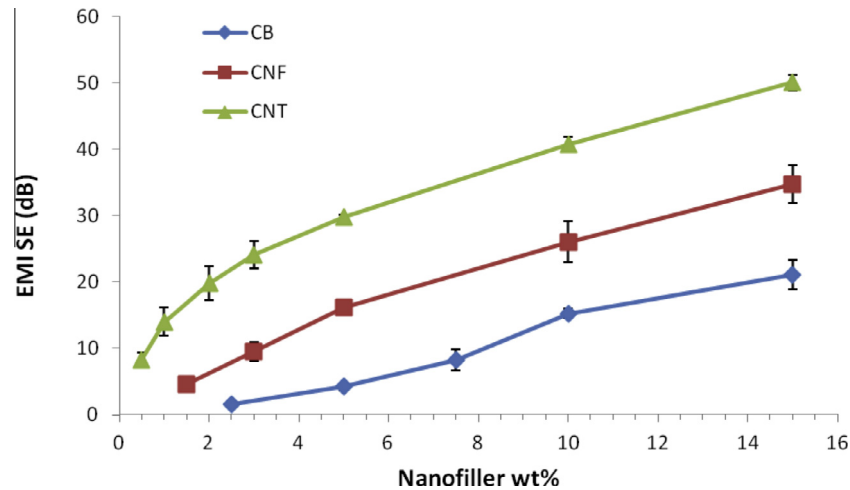


Fig. 4 – EMI SE at 8.2 GHz of different nanocomposites as function of nanofiller loading. Error bars represent the standard deviation.

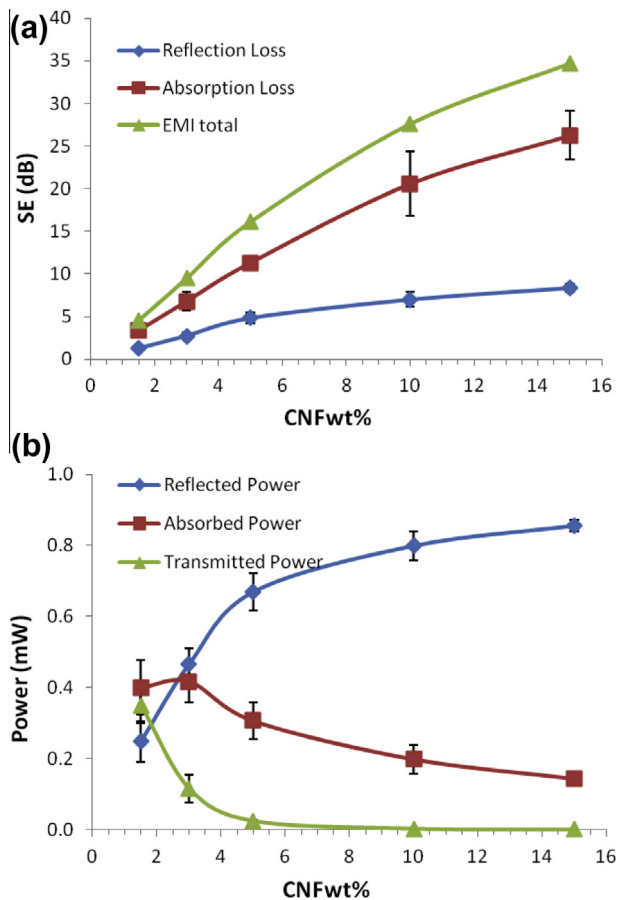


Fig. 5 – Shielding mechanisms (a) Absorption loss, reflection loss and total shielding as function of CNF content (b) power balance as function of CNF content.

The results in Fig. 5(a) show that absorption loss and reflection loss increase with the increase in nanofiller concentration (i.e. as the nanocomposite conductivity increases). Regardless of the CNF loading, absorption loss is higher than the reflection loss. For example, for the 10 wt.% CNF/ABS

nanocomposite, absorption loss is 20.6 dB compared to a reflection loss of 7.0 dB giving an overall EMI SE of 27.6 dB. The power balance presented in Fig. 5(b) clearly shows a remarkable increase in the reflected power and decrease in the transmitted and absorbed powers with the increase in the nanocomposite conductivity. For the 15 wt.% nanocomposite, the reflected power is 86% of the incident power. By comparing the results in Fig. 5(a) and Fig. 5(b), it can be concluded that nanocomposite materials have high intrinsic absorption capabilities. However, since reflection takes place before absorption most of the incident wave (as an absolute value) is reflected. Thus, under the current conditions the nanocomposites are not useful for applications requiring low magnitude of power reflection. However, they would be useful for applications requiring high reflectivity; such as lateral guidance of vehicles on highways [1]. In order to increase the absorbed power, a multilayered composite material of graded electrical conductivity can be used.

3.5. Dielectric characteristics

To further investigate the EMI shielding characteristics of the nanocomposites, the complex permittivity of the nanocomposites was extracted from the S-parameters using Nicholson–Ross–Weir method [35,36]. The complex permittivity ($\epsilon^* = \epsilon' + i\epsilon''$) constitutes the real component (ϵ') and imaginary component (ϵ'') representing, respectively, the polarization loss and the electric loss [43]. Nanocomposites with high real and imaginary permittivity are expected to have high EMI SE [9]. Fig. 6 depicts the complex permittivity of the three different nanocomposites at 8.2 GHz as a function of nanofiller weight percent. Fig. 6(a) shows that all nanocomposites exhibited an increase in the permittivity with the increase in nanofiller loading. However, for CNF/ABS and CNT/ABS nanocomposites the increase in ϵ' with nanofiller loading is more remarkable compared to that of CB/ABS nanocomposite. Moreover, it is apparent that at high nanofiller loading, ϵ' of CNT/ABS nanocomposites is considerably higher than that of CB/ABS and CNF/ABS nanocomposites. Real permittivity of polymer nanocomposites is a measure of the number of

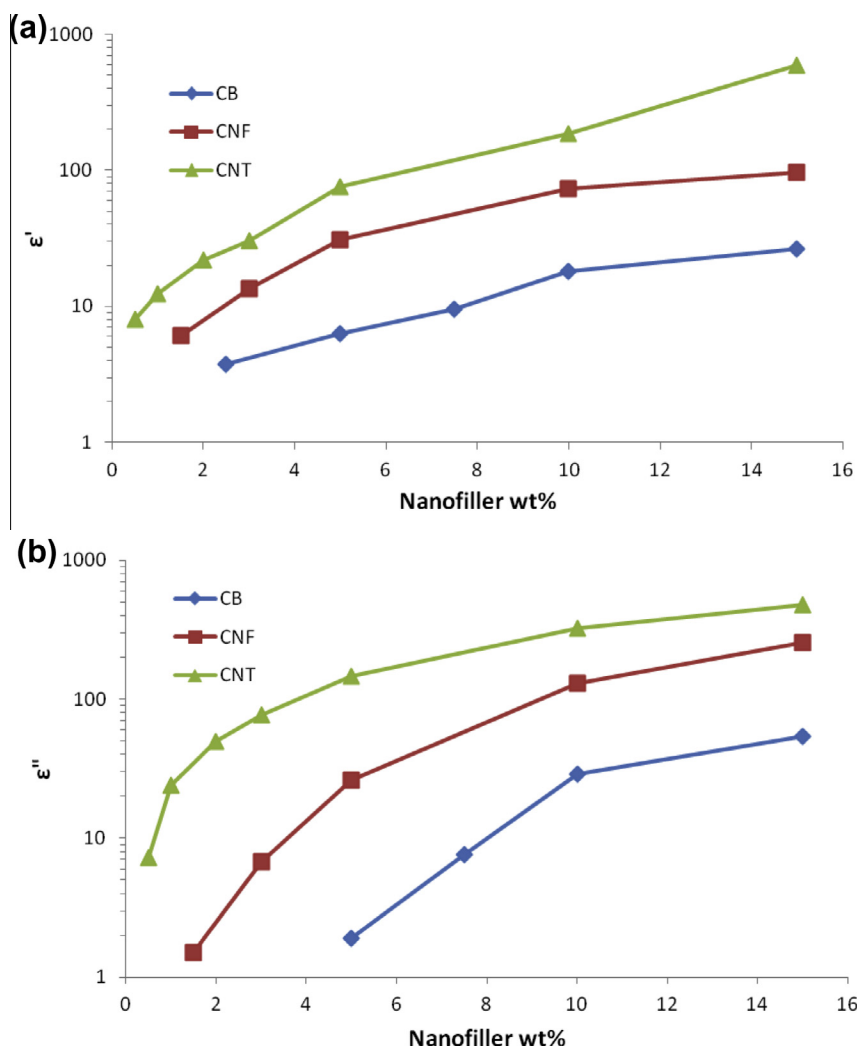


Fig. 6 – Real part (a) and imaginary part (b) of complex permittivity of the nanocomposites at 8.2 GHz as a function of nanofiller content.

micro-capacitors and the polarization centers [12,43]. Polarization centers originate from the defects in the nanofiller structure [43], while the micro-capacitors are simply formed by nanofiller particles/aggregates acting as electrodes filled with insulating polymeric material. Thus, the increase in CNF and CNT loading increases ϵ' because of the increase in the number of these micro-capacitors and structural defects. Moreover, the gap between the nanoparticles decreases as a result of increasing nanofiller loading, thus increasing the polarization of the polymeric material filling the gap between the nanoparticles; and consequently enhancing the absorption loss. Many studies reported that the real permittivity at 8 GHz of CNT/polymer nanocomposites containing around 15 wt.% CNT is in the range of 60–70 [9,12]. In this work, the real permittivity of the 15 wt.% CNT/ABS nanocomposite at 8.2 GHz is 590. The high relative permittivity reported in this work may indicate that the level of nanotubes dispersion in the ABS matrix is better than that in the aforementioned studies.

The imaginary permittivity of the nanocomposites is shown in Fig. 6(b). It is evident that the imaginary permittivity

increases with the increase in nanofiller loading. For 10 wt.% CNT/ABS nanocomposite, the imaginary permittivity is 323. This value is higher than that ($\epsilon'' = 150$) reported for CNT/polystyrene nanocomposite [12]. These difference might be attributed to the better dispersion of the CNT in the ABS matrix compared to that in the PS matrix and/or to the higher intrinsic conductivity of the CNT used in this study compared to CNT used in the other study [12]. In nanocomposites, conductive nanofiller networks act as dissipating mobile charge carriers. The increase in the number of conductive networks, as a result of nanofiller addition, leads to higher imaginary permittivity and consequently higher electromagnetic radiation dissipation by absorption. At any given nanofiller content, the imaginary permittivity of the nanocomposites has the following order: CNT/ABS > CNF/ABS > CB/ABS. The ratio of the imaginary permittivity of CNT based nanocomposite to that of CNF and CB nanocomposites, at any given nanofiller loading, decreases with the increase in the nanofiller concentration. For example, at 5 wt.%, the imaginary permittivity of CNT/ABS nanocomposites was 76 times that of CB/ABS nanocomposite and 5 times that of CNF/ABS nanocomposite.

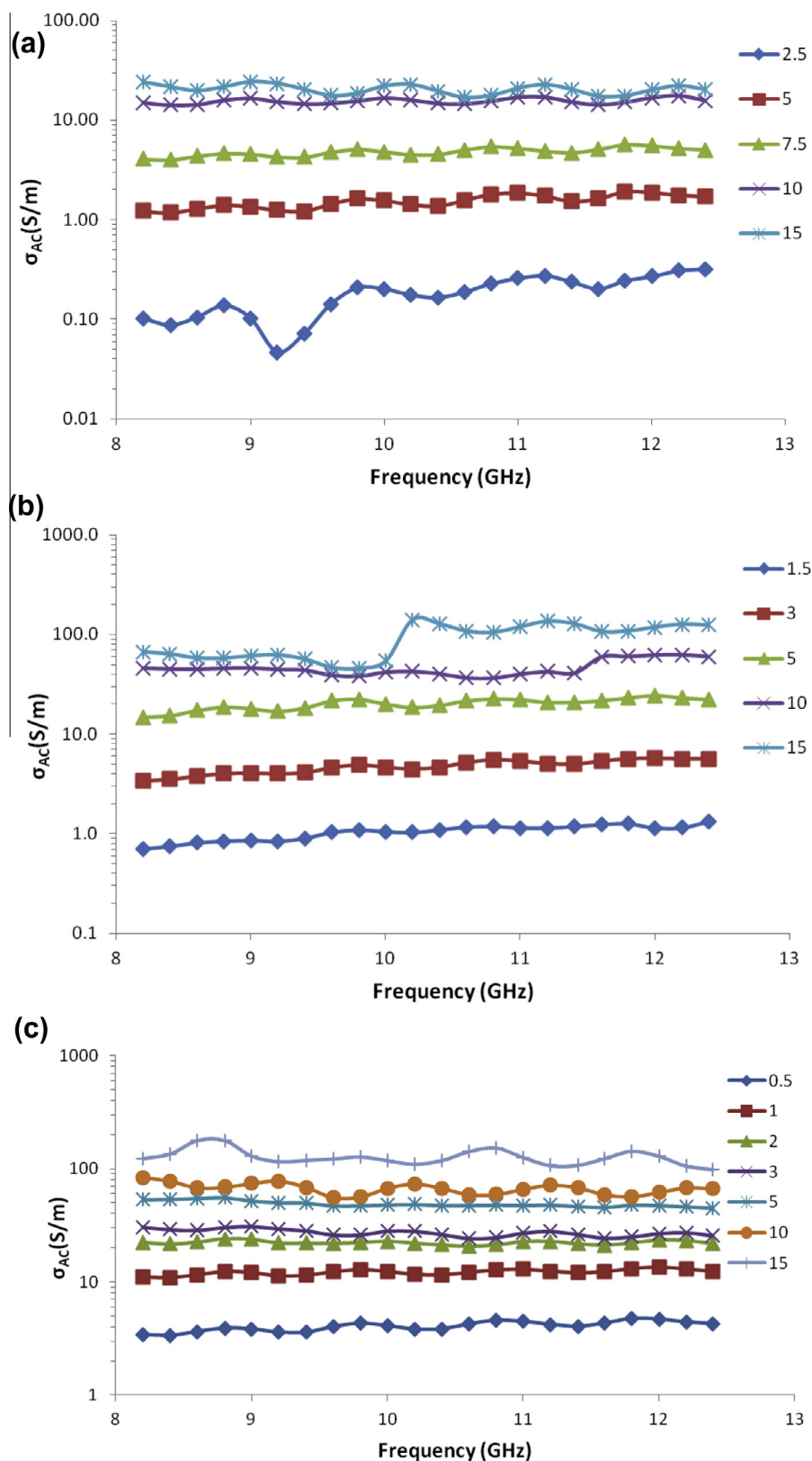


Fig. 7 – AC conductivity of (a) CB/ABS nanocomposite, (b) CNF/ABS nanocomposite and (c) CNT/ABS nanocomposite in the X-band frequency range.

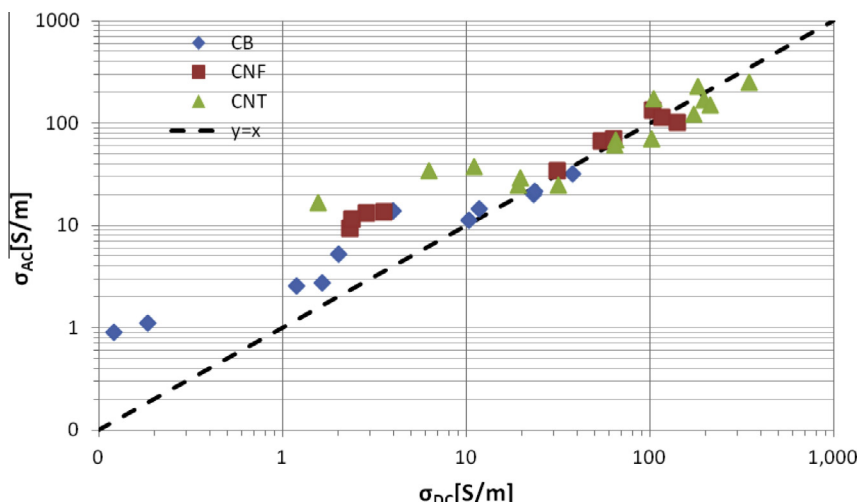


Fig. 8 – Relation between DC conductivity and AC conductivity at 8.2 GHz.

However, at 15 wt.%, the imaginary permittivity of CNT/ABS nanocomposites was 9 and 2 times, respectively, that of CB/ABS and CNF/ABS nanocomposites.

3.6. AC conductivity vs. DC conductivity

The AC conductivity (σ_{AC}) of the nanocomposites can be calculated based on the imaginary part of permittivity and the frequency as follows:

$$\sigma_{AC} \text{ (S/m)} = 2\pi f \epsilon_0 \epsilon''$$

where, f is the frequency in Hz and ϵ_0 is the permittivity of free space ($\epsilon_0 = 8.854 \times 10^{-12}$ F/m). Fig. 7 depicts the AC conductivity of the nanocomposites as function of frequency and nanofiller loading. Regardless of the nanofiller type, two major observations can be drawn from the figure. The first observation is there is an increase in the AC conductivity with increasing the nanofiller concentration. The second observation is AC conductivity almost independence of frequency in the X-band range. Based on the second observation, we tried, as shown in Fig. 8, to find a relation between the AC conductivity at 8.2 GHz range and the DC conductivity measured by the 4-wire probe method. It is evident that the AC conductivity is remarkably higher than the DC conductivity for samples with $\sigma_{DC} < 1$ S/m, i.e. for samples with low nanofiller loading where hopping of electrons can contribute to the nanocomposite conductivity. However, at higher nanofiller concentration, more conductive networks are built within the polymer matrix and the AC conductivity approaches the DC conductivity.

4. Conclusion

The EMI SE, electrical and dielectric properties of CNT/ABS, CNF/ABS and CB/ABS nanocomposites were studied to reveal the effect of nanofiller type on the aforementioned properties, to analyze the EMI shielding mechanisms, and to determine the relation between DC conductivity at the AC conductivity in the X-band frequency range. CNT/ABS nanocomposites exhibited outstanding characteristics compared to those of

CNF/ABS and CB/ABS nanocomposites. For example, at nanofiller loading of 5 wt.%, the EMI SE of CNT-based nanocomposite was 2 times that of CNF-based nanocomposite and 7 times that of CB-based nanocomposite. In addition, at CNT content of only 2 wt.%, the CNT/ABS nanocomposite had an EMI SE of 20 dB. Moreover, the results showed that CNT-based nanocomposite can be used in applications requiring high level of EMI attenuation; for example, a SE of 50 dB was reported for 15 wt.% CNT/ABS nanocomposite. Regardless of the nanofiller type, absorption loss was higher than reflection loss. However, because of the remarkable mismatch between the impedance of wave in free space and the impedance of the conductive nanocomposites, most of the incident power was reflected. AC conductivity was found to approach the DC conductivity at higher nanofiller concentration because of the increase in number of conduction pathways.

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